Ordinary Polynomial Multiplication over Every Fixed Finite Field

A strict logarithmic saving on fixed tapes

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Abstract

For each fixed finite field $\mathbb{F}_q$, ordinary polynomial multiplication has deterministic bit complexity $O_q(n(\log(n+2))^{1301/1302})$ on a fixed finite-tape machine. For each fixed odd field $\mathbb{F}_{p^e}$, the stronger bound $O_{p,e}(n(\log(n+2))^{319/320})$ follows from the cyclotomic-host reduction proved here. The characteristic-two branch uses the paired-phase binary theorem [4] and an explicit fixed-basis reduction costing a constant number of binary products plus linear coefficient conversion and recombination. Both branches return all $2n-1$ ordinary coefficients and charge setup, arithmetic, tape movement, output and cleanup. The common exponent is the larger of the two field-specific exponents; the sharper odd-field exponent is not asserted in characteristic two.

AI assistance and formal verification. Substantial generative AI assistance, including OpenAI Codex [20], supported mathematical exploration, algorithm and proof development, writing, coding, literature review, and source inspection. Historical model versions were not consistently recorded. The named author directs the project and makes publication decisions. The results are supported by the written arguments and named imports; complete Lean formalization and independent human mathematical review remain outstanding. Finite computational evidence is identified at its stated scope in the supplementary material.

1 Prior work and source lineage

For each fixed prime p, Harvey, van der Hoeven and Lecerf obtained $O_p(n\log n\, 8^{\log^* n})$ bit operations [6]. Harvey and van der Hoeven improved this to the unconditional $O_p(n\log n\, 4^{\log^* n})$ benchmark on deterministic fixed tapes and supplied the cyclotomic hosts used here [1]. Fixed-degree basis reduction extends that comparison to every individually fixed finite field. Their later uniform finite-field $O(n\log q\log(n\log q))$ theorem assumes a Linnik constant $L < 1+2^{-1162}$ [7, Theorem 1.2]. Harvey's October 2026 discussion identifies the fixed-field problem as a separate issue from the new integer-multiplication claims [8, Section 4]; it is context, not review of this work.

These developments established the algebraic and bit-complexity framework on which this paper builds. The $n\log n$ scale is a natural next benchmark for the unconditional fixed-field problem: the earlier uniform theorem reaches that scale with its stated number-theoretic assumption, while the unconditional cyclotomic algorithm retains the iterated-logarithmic factor. Our claims fix the field and address that remaining gap through the inherited hosts, new transfer accounting and accompanying binary constructions.

The source lineage is explicit. OpenAI's finite-network and tape interchange method [9] is an antecedent. The selected odd transfer imports the precise stage-two machine theorem [2], not an end-to-end integer multiplier. The native supplement retains community chart, PR6030 phase-word invoice [12], and contributor attributions [10]; related mixed-recursion and exact-complex RAM work is credited separately [11]. General fixed-matrix tensor acceleration in Alman's lookup-table model [15] is prior art, with a different access model. The supplement uses classical integer arithmetic [16, 17] and classical linear/quadratic Clifford descriptions [18]; its specific community-derived coordinate constructions retain their own credit. These classical and community developments provide essential antecedents to the construction.

The stronger interchange construction uses the distinct PR589 BIT record family at the pinned revision in [13]; it is not the raw-CU phase word. OpenAI's exact-complex DFT construction [14] is also a method-level antecedent for local kernel networks and Fourier layers. Its stated exact-complex arithmetic model is distinct from the finite-field bit model considered here. The present source comparison does not establish priority against every possible finite-characteristic adaptation.

The contribution recorded here is the odd-field host transfer and its combination with the accompanying binary construction, under the stated field and machine quantifiers. The companion baseline note [19] instead combines the accessible binary route with the odd proposition below to obtain $O_q(n \log(n+2))$. These are related results with shared foundations.

The odd proposition does not use the binary multiplication theorems or the combined all-field theorem. The paired-phase binary proof likewise does not invoke the combined conclusion. The composition below is therefore independent of that conclusion.

The odd-field construction uses the mixed-BIT and odd-ring phase arguments in the native supplement. Its direct quotient selector, three-call local word and charged native relocations provide the exponent 319/320. The characteristic-two construction supplies exponent 1301/1302 through Theorem 1 of [4]. The earlier controlled-CRT construction [5] remains a separate binary result; it is not a premise of the all-field composition in this paper.

2 Model and scope

The input consists of the ordinary coefficient lists of two polynomials of degree less than n over a fixed field $\mathbb{F}_q$, in a fixed polynomial basis. The output is the full ordinary coefficient list of their product. Time counts bit operations, including encoding, setup, movement, decoding, and output, on a deterministic machine with a fixed finite alphabet and a fixed finite number of sequential tapes. Constants may depend on the fixed field. No growing-field uniformity, practical cryptographic speedup, or combined workspace bound is claimed.

The logarithm is base two; changing its base changes only constants. All statements concern ordinary univariate multiplication. Group convolution, transformed products, integer multiplication, and multivariate products are different problems and are not used as output substitutes.

3 The odd-characteristic theorem

Proposition 1 (Odd-field ordinary product)**.** *For each fixed odd prime p and positive integer e, ordinary degree-$< n$ polynomial multiplication over $\mathbb{F}_{p^e}$ is deterministic and takes*

$$O_{p,e}\Big(n(\log(n+2))^{319/320}\Big)$$

bit operations on a fixed finite-tape machine.

Construction and dependency closure. Appendix A gives the current complete transfer. Its native inputs are the equal-field interchange and odd-ring phase theorems in `proofs/stronger-odd-20261102/native-source.tex.txt`. The relevant labels are `thm:mixedbitswap` and `thm:mixedbitinteger`. The supplement states the phase conclusion separately from the original integer application. The named external machine input is the stage-two interchange theorem at the pinned source in [2]. The new source supplies its additional finite-word, integer-flow, precompilation and delayed-cover arguments. These are written proof dependencies, not inferences from finite test counts.

For fixed p, the unconditional cyclotomic host [1] gives $N = \Theta(n)$ with $N > 2n-1$. Enlarge the conductor only after fixing the generator degrees. With $L = \lceil \log_2 N \rceil$ and $D = \varphi(\alpha)$, the literal coefficient algebra

$$B = \mathbb{F}_p[I, U]/(I^2 + 1, \Phi_\alpha(U))$$

has D Gaussian lanes. Its p^D-power Frobenius fixes both generators; thus $f = \mu^{p^D-1}$ is an exact idempotent with $(1 - f)\mu = 0$ for every local coefficient μ. Explicit canonical-U unit witnesses give a three-C word for every required shear, including zero divisors. The Fourier factorization and arbitrary-dirty exact-width printer retain every port and dummy until their completed boundaries.

Native aggregate relocations are repriced with the proved lossless binary-transpose corollary, rather than the older shuffle exponent. Diagonal table construction and application, both coefficient CRT components, the normalized inverse and ordinary reconstruction are separately charged. Small ring products use independent classical arithmetic [16, 17]. The resulting prime-field cost is

$$O_p\Big(n(\log(n+2))^{\gamma}(\log\log(n+4))^{K}\Big), \qquad \gamma = 1 - \frac{3125805}{10^9} = 0.996874195.$$

Since $319/320 - \gamma = 805/10^9 > 0$, the fixed secondary power is absorbed. A fixed basis uses e^2 serial prime-field products and $O_{p,e}(n)$ conversion and recombination work, returning all $2n-1$ coefficients. The proof and source map are in `proofs/stronger-odd-20261102/`; the older transfer sources remain historical support for version 0.1. □

Lemma 1 (Fixed binary-basis extension reduction)**.** *If ordinary multiplication over $\mathbb{F}_2$ on degree-$< n$ inputs has bit cost $T(n)$, then for each fixed $s \geq 1$, ordinary multiplication over $\mathbb{F}_{2^s}$ in a fixed polynomial basis has bit cost $O_s(T(n) + n)$ on the same fixed-tape model.*

Proof. Fix an $\mathbb{F}_2$-basis $\omega_1, \ldots, \omega_s$ of $\mathbb{F}_{2^s}$ and its structure constants before n varies. Write the input coefficient sequences as $f = \sum_{i=1}^{s} \omega_i f_i$ and $g = \sum_{j=1}^{s} \omega_j g_j$, where $f_i, g_j \in \mathbb{F}_2[X]$ have degree less than n. A sequential scan of the fixed-width coefficient encodings produces these $2s$ binary streams in $O_s(n)$ time. All basis indices and constants $c_{ijk} \in \mathbb{F}_2$, defined by $\omega_i\omega_j = \sum_k c_{ijk}\omega_k$, are fixed finite-control data. Initializing any working copies of their tables has constant field-dependent cost.

Execute the binary multiplier serially on fresh operand copies for each of the s^2 pairs (f_i, g_j). Each call returns all $2n-1$ ordinary product bits, including zeros. Scan that product into the s coordinate accumulator streams, adding it to coordinate k exactly when $c_{ijk} = 1$. Fixed-width encoding and decoding, operand copying, accumulator initialization, the $O(s^3 n)$ additions, temporary-stream erasure and head returns together cost $O_s(n)$. The accumulator streams are scanned coefficient by coefficient to write all $2n-1$ output coefficients in the prescribed basis and return the output head to its first symbol.

The binary calls charge their own arithmetic and tape movement. If a supplier does not already restore its private work tapes, mark their visited intervals on a fixed number of auxiliary tracks. Updating the endpoint marks has constant simulation overhead; clearing and resetting the intervals after a call takes $O(T(n) + n)$ time, since only that many cells

can have been visited. This normalization does not assume linear-size binary scratch. The paired-phase supplier [4] already includes its workspace cleanup in its stated bound. The number of streams, tracks, tapes and serial calls is fixed once s and the supplier machine are fixed. The total cost is therefore $O_s(T(n) + n)$ on one fixed finite-alphabet, fixed finite-tape machine, as claimed. □

Corollary 1 (Fixed characteristic-two fields)**.** *For each fixed $s \geq 1$, ordinary degree-$< n$ polynomial multiplication over $\mathbb{F}_{2^s}$ in a fixed basis takes $O_s(n(\log(n+2))^{1301/1302})$ deterministic bit operations on a fixed finite-tape machine, returning all $2n-1$ coefficients.*

Proof. Apply Lemma 1 to Theorem 1 of [4]. Since $1301/1302 > 0$, the linear wrapper cost is absorbed in the displayed bound. □

Theorem 1 (All-fixed-field bound)**.** *For every fixed finite field $\mathbb{F}_q$, ordinary degree-$< n$ multiplication in a fixed polynomial basis takes*

$$O_q\Big(n(\log(n+2))^{1301/1302}\Big)$$

deterministic bit operations on a fixed finite-tape machine.

Proof. Write $q = p^e$. If $p = 2$, Corollary 1 gives the stated bound using the paired-phase binary theorem and the fixed-basis reduction. If p is odd, Proposition 1 gives the stronger exponent $319/320$. Since $1301/1302 - 319/320 = 491/208320 > 0$, its logarithmic factor is bounded by the one displayed here for $n \geq 2$; finitely many smaller lengths are absorbed in the field-dependent constant. Both branches use ordinary coefficient representations and the fixed finite-tape bit-time model. The field and basis are fixed before n varies. The machine, tape count and constants may depend on them but are independent of n. Neither branch invokes the combined all-field conclusion. □

Theorem 2 (Unconditional fixed-field baseline)**.** *Every fixed finite field has an unconditional ordinary multiplication algorithm in $O_q(n \log(n+2))$ bit operations on the stated machine.*

Proof. This follows immediately from Theorem 1, because $(\log(n+2))^{1301/1302} = O(\log(n+2))$. The independent Pascal-transform route for characteristic two is also recorded in [3]. The odd fields follow from Proposition 1. □

4 Verification and limitations

The supplementary material supplies the native finite compiler, its recurrence, the cover-reserve calculation, and the finite native records. Saved finite-check reports support their stated finite identities; the asymptotic claims use the written proofs. The characteristic-two claim is imported from the paired-phase binary preprint [4], whose result retains its own proof scope. Lemma 1 charges its fixed-extension wrapper; the all-field theorem combines that corollary with the odd proposition. This composition is a consequence of the two written component theorems, not a new inference from finite computations.

Complete Lean proofs of the odd proposition and the combined theorem remain unfinished. The finite evidence and checked supporting components do not establish the complete asymptotic algorithms.

A Stronger odd-field transfer

This appendix gives the odd-characteristic argument with its coefficient quotient, root interface, native service tariffs, and ordinary input and output accounting. The non-elementary native services are named source theorems with written construction proofs in the native supplement described below.

Proposition 2 (Odd-field ordinary product at the stronger exponent)**.** *For each fixed odd prime p and fixed integer $e \geq 1$, there is a deterministic machine with a fixed finite alphabet and a fixed finite number of sequential tapes which multiplies two ordinary degree-$<n$ polynomials over $\mathbb{F}_{p^e}$ in a chosen fixed basis and outputs all $2n-1$ ordinary coefficients, including zero leading or trailing coefficients. Its bit complexity is*

$$\mathcal{O}_{p,e}\Big(n(\log(n+2))^{\gamma}(\log\log(n+4))^{K}\Big), \qquad \gamma = 0.996874195,$$

for a fixed constant K. Consequently it is

$$\mathcal{O}_{p,e}\Big(n(\log(n+2))^{319/320}\Big).$$

The field, basis, constants, and finite machine are fixed before n varies. No growing-field or uniform-in-e bound is asserted.

Proof. We first construct the machine over $\mathbb{F}_p$. The fixed-extension reduction is given at the end. All logarithms are base two.

A.1 Cyclotomic host and the ordinary coefficient maps

Fix a sufficiently large n; the finitely many smaller values use a fixed schoolbook branch. The unconditional cyclotomic construction of Harvey and van der Hoeven supplies a squarefree integer N with

$$2n < N < 4n, \qquad p \nmid N,$$

its prime factor tuple, and principal-root data [1]. Write

$$L_0 = \lceil \log_2 N \rceil, \qquad a = \lceil \log_2 L_0 \rceil, \qquad R_0 = 2^{a^2}.$$

Every prime factor q_i of N lies between L_0^3 and R_0 once n is large enough. The target required by the cyclotomic construction exceeds its fixed p^2 threshold; the direct branch handles the remaining inputs.

The public algorithms for local shapes, permutations, and route settings are fixed first, with a fixed effective polynomial bound on their record work. Starting with the conductor α_0 supplied with the root, append unused prime factors of N to obtain a squarefree conductor α for which

$$D = \varphi(\alpha) \geq (R_0 + L_0 + 5)^A$$

for a fixed sufficiently large A. Each appended factor contributes $q_i - 1 > L_0^3/2$, so $O(\log L_0)$ factors suffice. With

$$m = N/\alpha, \qquad \theta = 1 - \frac{D}{\alpha},$$

the conductor and the remaining host obey

$$\log \alpha, \log D = O((\log L_0)^4), \quad \theta = O((\log L_0)^2/L_0^3), \quad mD = N(1-\theta) = \Theta(N). \tag{1}$$

The number of prime axes remaining in m is large enough for the native dock: the original tuple has at least $(L_0 - O(1))/\log_2 R_0$ axes, and deleting the $O((\log L_0)^2)$ conductor factors still leaves the required

$$h = \left\lfloor \frac{L_{\mathrm{dock}}}{40\lceil \log_2(R_0+1)\rceil} \right\rfloor, \qquad L_{\mathrm{dock}} = \lceil \log_2(2M)\rceil, \qquad M = \prod_{q_i \mid m} (q_i + 1).$$

Here $L_{\mathrm{dock}} = L_0 - O((\log L_0)^4) = \Theta(L_0)$ and

$$\frac{M}{m} = \prod_{q_i \mid m} \left(1 + \frac{1}{q_i}\right) = 1 + O(L_0^{-2}), \qquad MD = \Theta(N).$$

The large coefficient degree D pays the public setting-generation cost; the separate relation $MD = \Theta(N)$ pays the data volume.

Let

$$S = \mathbb{F}_p[U]/(\Phi_\alpha(U)), \qquad A_p = \mathbb{F}_p[I]/(I^2+1), \qquad B = A_p[U]/(\Phi_\alpha(U)).$$

These are literal monic quotients. The ring S has D scalar coordinates, while B has D lanes over A_p, or $2D$ scalar coordinates over $\mathbb{F}_p$. Neither quotient is assumed to be a field.

There are two distinct coefficient maps in the construction. First, ordinary input coefficients use the new parameter map $Y_{\mathrm{new}} \mapsto U$. Separately, the old principal root is transported from the original host: the host parameters satisfy $q_0 \mid \alpha_0 \mid \alpha \mid N$. Put $J = \alpha/\alpha_0$. Since α is squarefree, $\gcd(J, \alpha_0) = 1$, and the cyclotomic identity

$$\Phi_{\alpha_0}(X^J) = \prod_{d \mid J} \Phi_{\alpha_0 d}(X)$$

implies $\Phi_\alpha(X) \mid \Phi_{\alpha_0}(X^J)$. Consequently $U_0 \mapsto U^J$ defines a unital homomorphism φ from the original coefficient quotient to S. If ρ_0 has principal order N/q_0 in the original quotient, set

$$\omega_0 = \rho_0^{\alpha_0/q_0}, \qquad \omega = \varphi(\omega_0)^J, \text{where } \varphi : U_0 \longmapsto U^J.$$

Taking the indicated divisor powers gives principal orders N/α_0 and $m = N/\alpha$, respectively. The homomorphism preserves the vanishing character sums defining principality. Root transport uses this homomorphism separately from the ordinary-input coefficient map.

The cyclic coefficient algebra of length $N = \alpha m$ is represented by

$$\mathbb{F}_p[X]/(X^N - 1) \simeq \mathbb{F}_p[U, Z]/(U^\alpha - 1, Z^m - 1), \qquad X \longmapsto U^c Z, \quad cm \equiv 1 \pmod{\alpha}.$$

For $k = qm + r$, the monomial X^k maps to $U^{q + cr \bmod \alpha} Z^r$. The forward map, its inverse, the short α-direction transpose, and row rotations cost $O_p(N(1 + \log \alpha))$ on fixed banks. This is the ordinary coefficient map, not the root lift above.

Both coefficient CRT components are retained. Since $p \nmid \alpha$, $U^\alpha - 1$ is squarefree over $\mathbb{F}_p$, so

$$U^\alpha - 1 = \Phi_\alpha(U)\, \psi(U), \qquad \psi(U) = \frac{U^\alpha - 1}{\Phi_\alpha(U)}, \qquad \gcd(\Phi_\alpha, \psi) = 1.$$

Extended Euclid prepares the two CRT idempotents and inverses. The ψ-component is reconstructed along with the Φ_α-component; discarding it would lose ordinary input information.

A.2 The actual quotient selector and its unit witnesses

The algebraic selector is proved inside the actual quotient B. The conductor α is odd and greater than one, so $D = \varphi(\alpha)$ is even. Put $Q = p^D$. Euler's theorem and oddness give

$$Q \equiv 1 \pmod{\alpha}, \qquad Q \equiv 1 \pmod{4}.$$

In B, $U^\alpha = 1$ and $I^2 = -1$, hence $U^Q = U$ and $I^Q = I$. Frobenius to the D-th power is a ring homomorphism in characteristic p and fixes $\mathbb{F}_p$. Every element has the monic-basis expression

$$x = \sum_{j=0}^{D-1} (a_j + b_j I)U^j, \qquad a_j, b_j \in \mathbb{F}_p,$$

so, by applying Frobenius to the generators and coefficients,

$$x^Q = x \quad \text{for every } x \in B. \tag{2}$$

This uses the quotient generators themselves. It does not factor B or identify it with a field.

For each local coefficient $\mu \in B$, define

$$f = \mu^{Q-1}, \qquad r = 1 - f.$$

Equation (2) gives

$$f^2 - f = \mu^{Q-2}(\mu^Q - \mu) = 0, \qquad f\mu = \mu^Q = \mu, \qquad r\mu = 0. \tag{3}$$

Thus f and r are complementary orthogonal idempotents. In particular, the nilpotent remainder needed by the generic word has bound one; the fixed-field argument uses no Hensel or Newton iteration.

Put $h = 2^{-1} \in \mathbb{F}_p$ and $q_U = U^2 + 1$. The units used below have explicit inverses in S. Certainly $U^{-1} = U^{\alpha-1}$. For $U - 1$, let

$$c_\alpha = \Phi_\alpha(1), \qquad H(X) = \frac{\Phi_\alpha(X) - c_\alpha}{X - 1}.$$

Then $(U-1)H(U) = -c_\alpha$. Since α is squarefree, c_α is either 1 or the prime factor of α when α is a prime power; either value is nonzero modulo p. Hence $(U-1)^{-1} = -c_\alpha^{-1}H(U)$. For $j = 1, 2$, oddness of α gives

$$(1 + U^j)\sum_{k=0}^{\alpha-1}(-U^j)^k = 1 - (-U^j)^\alpha = 2,$$

so

$$(U+1)^{-1} = h\sum_{k<\alpha}(-U)^k, \qquad q_U^{-1} = h\sum_{k<\alpha}(-U^2)^k.$$

These sums are prepared once by geometric doubling; no inverse is taken as an algebraic oracle.

The shared canonical parameter and all required inverses are

$$\begin{aligned} u &= fU + rI, & u^{-1} &= fU^{-1} - rI, \\ v_- &= f(U-1)^{-1} - r(I+1)h, & (u-1)v_- &= 1, \\ v_+ &= f(U+1)^{-1} + r(1-I)h, & (u+1)v_+ &= 1, \\ c &= fq_U\mu^{Q-2} + r, & c^{-1} &= f\mu q_U^{-1} + r. \end{aligned}$$

Orthogonality of f, r, together with (3), proves each displayed inverse identity. In particular,

$$u^2 + 1 = fq_U = c\mu, \qquad cc^{-1} = 1. \tag{4}$$

The unit c is explicit even when μ or $u^2 + 1$ is zero or a zero divisor. No factorization, zero test, or residue-field conversion is used.

A.3 The exact three-C shear word

Write $i = I$, $A = (1+i)h$, and $B_0 = (1-i)h$, and set

$$C = \begin{pmatrix} A & B_0 \\ B_0 & A \end{pmatrix}.$$

Then $AB_0 = h$, $A^2 = ih$, $B_0 = -iA$, and $2AB_0 = 1$. Let

$$b = (u-1)v_+, \quad \alpha_1 = 2B_0uv_+, \quad \beta_1 = A(u^2+1)v_+, \quad \delta_1 = B_0(u-1).$$

Direct multiplication, using $(u+1)b = u-1$, gives the literal three-call core

$$C\begin{pmatrix} u & 0 \\ 0 & 1 \end{pmatrix} C \begin{pmatrix} b & 0 \\ 0 & 1 \end{pmatrix} C = \begin{pmatrix} \alpha_1 & \beta_1 \\ 0 & \delta_1 \end{pmatrix}. \tag{5}$$

For example,

$$C\begin{pmatrix} u & 0 \\ 0 & 1 \end{pmatrix} C = \begin{pmatrix} ih(u-1) & h(u+1) \\ h(u+1) & -ih(u-1) \end{pmatrix};$$

right multiplication by $\mathrm{diag}(b,1)C$ gives (5). Thus the identity is a product of three positive C factors, not a matrix supplied as a premise.

The two nonzero diagonal pivots in (5) have the following explicit inverse witnesses:

$$\alpha_1^{-1} = A(u+1)u^{-1}, \qquad \delta_1^{-1} = 2Av_-.$$

Indeed $\alpha_1\alpha_1^{-1} = 2AB_0(uv_+)(u+1)u^{-1} = 1$ and $\delta_1\delta_1^{-1} = 2AB_0(u-1)v_- = 1$. Normalize the left side of the core by these units. The resulting upper shear has parameter

$$\kappa = \alpha_1^{-1}\beta_1 = A^2(u^2+1)u^{-1} = ih\,c\mu u^{-1}.$$

Now put

$$z = -2iuc^{-1}, \qquad z^{-1} = ihu^{-1}c.$$

Equation (4) gives $zz^{-1} = 1$ and $z\kappa = \mu$. Consequently the four diagonal factors

$$D_0 = \mathrm{diag}(z\alpha_1^{-1}, \delta_1^{-1}), \quad D_1 = \mathrm{diag}(u,1), \quad D_2 = \mathrm{diag}(b,1), \quad D_3 = \mathrm{diag}(z^{-1},1)$$

are all unit diagonals, and

$$D_0CD_1CD_2CD_3 = \begin{pmatrix} 1 & \mu \\ 0 & 1 \end{pmatrix}. \tag{6}$$

This one word works for every μ, including 0 and zero divisors. Its three calls and four diagonal slots have fixed topology. The generic identity is also the content of the arbitrary-coefficient theorem `arbitrary_threeC_word_correct` in `OddGeneralThreeCWord.lean`; the displayed host formulas supply its unit witnesses rather than assuming them.

A.4 Principal-root differences and the local transforms

The cyclotomic host supplies a principal root, whose defining character sums give explicit differences that are units. For any z in a commutative ring define

$$G_k(z) = \sum_{j=0}^{k-1} z^j, \qquad J_k(z) = \sum_{j=0}^{k-1} (k-j)z^j.$$

Telescoping (or induction using $J_{k+1} = J_k + G_{k+1}$) gives

$$(1 - z)J_k(z) = k - zG_k(z). \tag{7}$$

If $G_k(z) = 0$ and k is invertible, then $J_k(z)k^{-1}$ is an explicit inverse of $1 - z$. For a principal k-th root η, apply this to $z = \eta^r$ for $0 < r < k$. This argument works in product rings and requires no cancellation. Since all the lengths here divide m and $p \nmid m$, the needed scalar inverses exist. This identity proves unit existence for every divisor length of m; it does not evaluate a length-m sum for each record. The actual local transforms have prime lengths $q_i \le R_0$. For each required local root difference, direct evaluation of J_{q_i} uses a fixed polynomial number of operations in R_0 over B; an extended-Euclid calculation modulo Φ_α is another charged method with a fixed polynomial cost in D. These preparations are made in public setup, so the total is $n^{o(1)}$. No length-m sum is run per local record.

For each prime $q_i \mid m$, let

$$E_i = \frac{m}{q_i}\left(\left(\frac{m}{q_i}\right)^{-1} \bmod q_i\right), \qquad \zeta_i = \omega^{E_i}.$$

These are the coupled CRT idempotents and roots; they are derived from the one transported root rather than supplied by an independent root oracle. For $0 \le j < q_i$, put

$$H_j = \prod_{a=1}^{j}(1 - \zeta_i^a), \qquad H_0 = 1, \qquad \beta_j = (-1)^j \zeta_i^{j(j-1)/2}.$$

Equation (7) proves each H_j is a unit. The ring-valid Newton-basis identity factors the q_i-point Fourier matrix $F_{r,s} = \zeta_i^{rs}$ as

$$F = \mathcal{A}T \operatorname{diag}(\beta_j H_j^{-1}) T^{\mathsf{T}} \mathcal{A}, \qquad \mathcal{A} = \operatorname{diag}(H_j),$$

where T is unit lower triangular Toeplitz with defining sequence H_j^{-1}. For the entrywise proof, define

$$P_0(X) = 1, \qquad P_k(X) = \prod_{a=0}^{k-1}(X - \zeta_i^a), \qquad g_{j,k} = H_j H_k^{-1} H_{j-k}^{-1} \quad (0 \le k \le j < q_i).$$

Set $g_{j,k} = 0$ outside this range. The relation $XP_k = P_{k+1} + \zeta_i^k P_k$ and

$$g_{j,k-1} + \zeta_i^k g_{j,k} = H_j H_k^{-1} H_{j+1-k}^{-1}\big((1 - \zeta_i^k) + \zeta_i^k(1 - \zeta_i^{j+1-k})\big) = g_{j+1,k}$$

for $j + 1 < q_i$, with the two edge coefficients equal to one, prove by induction the formal polynomial identity $X^j = \sum_{k=0}^{j} g_{j,k} P_k(X)$ for every $j < q_i$. If $r < k$ then $P_k(\zeta_i^r) = 0$. If $r \ge k$, direct factorization gives

$$P_k(\zeta_i^r) = \prod_{a<k} \zeta_i^a(\zeta_i^{r-a} - 1) = \beta_k H_r H_{r-k}^{-1}.$$

Substitution in the polynomial identity yields

$$\zeta_i^{rs} = H_r H_s \sum_{k=0}^{\min(r,s)} \beta_k H_k^{-1} H_{r-k}^{-1} H_{s-k}^{-1},$$

which is exactly the (r, s) entry of the displayed factorization. Only H_j for $j < q_i$ is inverted; no H_{q_i} inverse, polynomial equality from finite evaluations, or field cancellation is used. The character sums give the inverse transform after multiplication by the genuine scalar q_i^{-1}.

For the local formal convolution let $\mathcal{R}$ be a nonzero commutative ring with a specified inverse h of 2; the actual coefficient ring B has this property. At length $v \geq 256$, choose b to be the least power of two with $b^2 \geq 4v$, put $s = b/2$ and $a = \lceil v/s \rceil$, and choose t to be the least power of two at least $2a - 1$. For $v < 256$ use direct multiplication. Since $b^2 \geq 4v$, $a \leq b/2$ and $t \leq b$. Minimality of b gives $b < 4\sqrt{v}$. Also $t < 2(2a-1) < 4a \leq 8v/b+4$, whence $tb < 8v + 4b < 9v$ because $4b < 16\sqrt{v} \leq v$ for $v \geq 256$. Finally $b < 4\sqrt{v} \leq v^{3/4}$ in this range.

Work in the free monic quotient $\mathcal{C}_b = \mathcal{R}[Y]/(Y^b + 1)$ with generator y. Its free basis $1, y, \ldots, y^{b-1}$ shows $y^j \neq 1$ for $0 < j < b$; for $b \leq j < 2b$, $y^j = -y^{j-b} \neq 1$ by the same basis independence and $2 \in \mathcal{R}^\times$. Thus y has exact order $2b$. Since $t \leq b$ and both are powers of two, $t \mid b$, so $\omega = y^{2b/t}$ has exact order t. For $0 < r < t$, write $g = \gcd(r, t)$ and $d = t/(2g)$. Then r/g is odd and $(\omega^r)^d = y^{b(r/g)} = -1$, so

$$(1 - \omega^r) \left(h \sum_{j=0}^{d-1} \omega^{rj} \right) = 1.$$

Every nontrivial root difference is therefore an explicit unit, and t^{-1} is a unit because t is a power of two. If $t = 2^j$, prepare $t^{-1} = h^j$ and count each of its applications as a scalar gate. The radix-two transform and its inverse are valid over $\mathcal{C}_b$ with this genuine normalization. Split the input polynomials into width-s blocks. Their outer block product has degree at most $2a - 2 < t$, and each inner block product has degree at most $2s - 2 = b - 2$. Thus neither transform coordinate wraps; overlap-add recovers the exact product in $\mathcal{R}[X]$ and costs $O(ab) = O(v)$.

Counting additions, products and scalar operations, the recursive supplier has one length-b product for each outer transform point. The length-t transforms on t elements of $\mathcal{C}_b$ use $O(tb \log(t+2)) = O(v \log(v+2))$ coefficient-ring operations; powers of y are signed coordinate rotations, and the inverse normalization uses the prepared scalar t^{-1}. Folding, padding, and overlap-add cost $O(v + ab + tb) = O(v)$. Therefore

$$T(v) \leq tT(b) + Kv \log(v + 2), \qquad tb < 9v, \quad b \leq v^{3/4}.$$

Since $b \leq v^{3/4}$, we have $\limsup_{v\to\infty} \log(b + 2)/\log(v + 2) \leq 3/4$. Choose a fixed η with $9(3/4)^{16} < \eta < 1$. Beyond a fixed threshold, the recursive coefficient $9(\log(b+2)/\log(v+2))^{16}$ is at most η. Choose C large enough to absorb the $O(v \log(v+2))$ toll into $(1-\eta)Cv \log^{16}(v+2)$; increasing C also covers the finite intervening values and the direct range. Hence $T(v) = O(v \log^{16}(v + 2))$. The transform depth sums to $O(\log(v + 2))$ along a recursive path: these parameters give $t < 8v/b + 4 = O(\sqrt{v})$, so each transform has depth $O(\log(v + 2))$, while recursive child logarithms shrink by a factor at most 3/4.

Fixing the second convolution vector yields a linear DAG over $\mathcal{R}$: evaluate the finite arithmetic circuit in $\mathcal{R}[\epsilon]/(\epsilon^2)$, replace a variable input by ϵx_j and the fixed kernel by its constant value, and take the linear component. This is a ring identity and uses no field interpolation. Replace every fanout by a binary copy tree. The copied DAG has size $O(v \log^{16}(v + 2))$ and depth $O(\log^2(v + 2))$.

The off-diagonal block in the lower-unit-Toeplitz recursion has a rank-three displacement representation. Each of its three terms uses two fixed-kernel convolutions, so one composite chunk DAG contains six convolutions, all intermediate vectors, copies, pads, summands, and outputs. If u is the parent interval and w the larger chunk width, each chunk's source and target lie in different halves of the original parent. The fitting rule counts its actual source and target ports and every such gate, including the intermediate vectors, and yields $w = \Omega(u/\log^{16}(u + 2))$. There are $O(\log^{32}(u + 2))$ ragged chunk pairs per recursion level. Signed dirty replay executes the DAG, adds its outputs to the target, undoes it, then executes the source-suppressed DAG and subtracts its outputs before undoing. The common dirty

contribution cancels, the target receives exactly the intended chunk map, and every borrowed cell returns to its incoming value. Fanout copy trees and the four replay traversals give $O(\log^2(u+2))$ matching layers per chunk; recursion over $O(\log(u+2))$ levels therefore uses $O(\log^{35}(u+2))$ rounds on the original parent coordinates. The fixed finite range where a composite chunk does not fit uses direct triangular-shear words. This printer and its setup lists have polynomial size in R_0; the coefficient arithmetic and record width are charged separately below.

A.5 Native services and repriced routing

Let V denote the whole physical bit volume of a call, including complete prefixes, gaps, suffixes, independent lanes, coefficient bits, initialization, head movement, and cleanup. The native source supplies the following named theorem interfaces:

$$I_\tau : \text{opaque equal binary-field interchange}, \qquad \tau = 1 - \frac{859173879734039}{250000000000000000},$$
$$P_\gamma : C^{\otimes k} \text{ over } A_P = (\mathbb{Z}/P\mathbb{Z})[I]/(I^2+1), \qquad \gamma = 1 - \frac{3125805}{10^9} = 0.996874195.$$

The I_τ interface permits arbitrary complete spectators and opaque payload and costs $O(V(u+1)^\tau \operatorname{polylog}(u+2))$ for u-bit address fields. The P_γ interface applies the literal tensor over every odd modulus P, every coefficient width s, and every independent lane count ℓ at cost

$$O\Big(2^k \ell s (k+1)^\gamma \operatorname{polylog}(k+2)\Big).$$

Both tariffs include headers, record encodings, movement, all lanes, complete spectators, returns, and cleanup. The interchange proof is `thm:mixedbitswap`. The phase proof is `thm:mixedbitinteger`. Their construction uses the pinned source theorem `Stage2Machine.Bound.uniform_bound` [2]. The precise external statement and source revision are recorded in the supplementary dependency map. The finite-ring construction retains the contributor attributions recorded in [10].

Every aggregate relocation in the strong transfer is charged through I_τ as follows. Suppose two moved objects are contiguous complete numeric ranges of sizes A and B, with $AB \le M$. Pad each whole range separately to a binary power; the combined padded volume is less than $4AB$. Apply the lossless binary transpose to this complete rectangle and crop only after its completed permutation. Its address width is at most $\lceil \log_2 M \rceil + 2$, so each direction costs

$$O(VL^\tau \operatorname{polylog}(L+2)), \qquad L = \lceil \log_2(2M) \rceil. \tag{8}$$

This route pads a whole rectangle once. The earlier row-by-row routing estimate with exponent $\rho = 1 - 3 \cdot 2^{-38}$ is larger than γ and does not give (8).

At the selected host shape, the native radices satisfy $L_0^3 \le q_i \le R_0$, and the per-record width is at least the fixed polynomial required by the public generators, since D was chosen after that polynomial was frozen. The native dock uses disjoint data, dirty auxiliary, and movement fields; its untouched suffix has at least $0.8L$ bits and pays the polynomial address and gauge work. The source supplies the dock in `lem:nativedock` and the phase in `prop:nativephase`. The shuffle and ragged routing proofs are `lem:nativeshuffle` and `lem:raggednative`. Controlled permutations use `prop:nativecontrolled`; coupled CRT levels use `prop:nativeCRT` and `lem:CRTphase`. The wider host condition $\log R_0 = O((\log L)^2)$ follows from these proofs and the stated dock hypotheses; it is not inferred from the narrower polynomial-radix statement alone.

The strong native bounds satisfy

$$\tau < \gamma, \qquad (1-\tau) - \frac{3125805}{10^9} = \frac{77722629734039}{250000000000000000} > 0.$$

The shuffle has $O(\log R_0)$ endpoint blocks, ragged routing has $O(\log R_0)$ global levels, and balanced CRT routing has $O(\log t)$ depths, where t is the number of axes. Replacing every aggregate move by (8) charges these as fixed secondary powers of $\log L$. The native phase uses P_γ; no integer-multiplier conclusion from that source is used.

Each odd q_i axis is completed to the even radix q_i+1. Complete each local edge matching by pairing its remaining positions, including the existing dummy coordinate; the remaining count is even. Give every such added pair the coefficient $\mu = 0$. Equation (6) then acts as the identity on arbitrary incoming values, with all three C calls and all four diagonal slots still present. The generated local permutation places pairs in canonical adjacent positions; a separately priced route-in and route-out implements this change of layout. Shorter axis schedules are padded to the common tensor schedule with these zero-weight identity words.

In a synchronized tensor round, the product of local words is three full P_γ calls and four product-diagonal slots. It is a product of operator words, not an expansion into exponentially many additive terms. Inside a word, live values can occupy a dummy or the I helper coordinate; only its completed shear boundary is used for restoration or cropping. At completed boundaries the zero shears restore every dummy-pattern class, and the shear over S restores the helper subspace. The controlled and CRT routes preserve those same classes at their completed boundaries. Thus no crop or helper projection is inserted inside a native word.

All local edge matchings use $O(L_0 R_0 \log^{35}(R_0+2))$ coefficient records. The selector exponent has $O(D \log p)$ bits. Root differences, selector powers, diagonals, archives, shape lists, route settings, and erasure are prepared once per local record, not once per global sample. Their total work is a fixed polynomial in R_0, D, L_0, hence $n^{o(1)}$. The total volume of one native call is

$$V = 2MD\,\lceil \log_2 p \rceil = \Theta_p(n).$$

Including the fixed powers of secondary logarithms, all native phase, route-in, route-out, and CRT work is therefore

$$O_p\Big(n(\log n)^\gamma (\log\log(n+4))^{K_1}\Big) \tag{9}$$

for a fixed K_1.

The native BIT precompiler fixes one moment parameter $t \in (0,1)$ for which its productive and total profiles satisfy $P_t \le T_t$; it does not require this inequality for every t. The application uses $t = 1 - 173/50000$ and $z = 1 - 3459/10^6 > t$. For rational $z = a/b$, its finite stopping test is the exact integer comparison $F(N)^b < B(N)^b N^a$. The induction sets $B(0) = 1$, $F(0) = 0$, and $\ell(0) = 0$. Rank-zero complements are identities and are omitted; positive-rank recipes have positive productive count. At height G, the empty-target class has at most J^G columns and is admissible because $m_{\mathrm{flow}} J^G \le m_{\mathrm{flow}} L^G \le n$. Thus a source exists and the normalization denominator is positive. The finite winning word is selected before the later input widths vary; it is not an input-sized advice table or a compiler oracle.

The separated base word has dimension 127^{1024} in `lem:finitebitflow`. Its all-column proof and strictification feed `thm:finitebitcompiler`, which selects the effective finite winner and its constants $N_0 \ge 2$, B_0, and F_0; the enormous instruction list is not materialized. Every paid leaf has rank one. At width n, a rank-one child has width $\lfloor n/N_0 \rfloor$; aside from nonrecursive old-supplier leaves, each fallback child is no wider, so every recursive child has width at most $\lfloor n/N_0 \rfloor \le n/2$. For the rank-one tree, a safe path allowance is

$$D_0 = 2(\lceil \log_2(E+1) \rceil + 1),$$

where E is the root digit parameter. This uses the actual half-width contraction of the compiled word.

Fix the finite owned/route inventory and choose one eligible odd address prime q after the winning word is frozen. Put

$$L_{\mathrm{cov}} = B_0 |\,\mathrm{GL}_{N_0}(\mathbb{F}_q)|, \qquad A_w = q^{N_0^2(w-1)}, \qquad W_w = L_{\mathrm{cov}} A_w, \qquad w = \lceil E^{1/250} \rceil.$$

Choose the least r_0 with $q^{2r_0} \geq W_w^{D_0}$, then increase it by fewer than w digits to r so that $w \mid E - r$. Because

$$\log_q W_w = \log_q L_{\mathrm{cov}} + N_0^2(w-1) = O_{N_0,B_0,q}(w),$$

we have $r = O_{N_0,B_0,q}(w \log(E+2)) = o(E)$. The finitely many E for which $r \geq E$ use a direct branch. Exchange the two high r-digit fields and place the right high field before the low head field; these short moves cost $O(Vr)$. Pad their joined q^{2r} row values once to

$$R' = W_w^{D_0} \left\lceil \frac{q^{2r}}{W_w^{D_0}} \right\rceil < 2q^{2r}.$$

The range supplies an outer factor and D_0 consecutive W_w cover slots. This is a grouped complete numeric range, not entrywise radix conversion.

A child fixes only a productive role and low matrix type, consuming one fresh cover slot while retaining all A_w high matrices as complete earlier spectators. At depth d, retained-high multiplicity A_w^d times unused stock R'/W_w^d is exactly R'/L_{cov}^d. Thus the child volume is divided by L_{cov}, not by W_w; there is no second high-matrix saving. Siblings reuse restored stock, parent inactive banks remain below their stack tops, and the finite owned-bank inventory travels with the active stream until its prescribed clearing boundary. Only the completed root endpoint restores the padded row support for cropping. This is the retained-stock argument of `lem:threecoverstock`, generalized to the winning finite word by `prop:delayedbitcover`.

The finite program gives

$$C_0 = \frac{3459}{10^6}, \qquad C_1 = \frac{69}{20000}, \qquad \frac{F_0}{B_0} < N_0^{1-C_0}, \qquad d = C_0 - C_1 = \frac{9}{10^6}.$$

Choose q large enough to avoid every finite denominator and rank witness and to satisfy $q > 256N_0^5/d$. With $\sigma = 1 - C_1$, the fresh-cover density and full fallback give

$$\mu_\sigma \leq \frac{F_0}{B_0} N_0^{-\sigma} \left(1 + \frac{64N_0^5}{q}\right) < N_0^{-d}(1 + d/4) < \frac{1 + d/4}{1 + d/2} < 1, \qquad \mu_1 \leq \mu_\sigma.$$

The last inequality uses $N_0^d \geq 1 + d/2$, since $N_0 \geq 2$ and $\ln 2 > 1/2$. This is the BIT density estimate. The separate phase moment theorem `lem:strongerphasemoment` supplies P_γ; it is not used for the BIT density bound.

The ancestor decoder is charged by `lem:rootdecoder`. Its descriptor work is polynomial in E and is absorbed by the complete q^w target fibres, because $E \leq w^{250}$. Let $a_L = 123879734039/10^{15}$ and $\tau_L = 1 - a_L$, so a port costs $G(w) = (w+1)^{\tau_L} \operatorname{polylog}(w+2)$ per active volume. The strict profile bound gives the recurrence estimate $T(n,w) \leq An + Bn^{1-C_1} G(w)$, with the finite generated-adapter cost charged linearly. The one-reserve wrapper costs

$$O\Big(V\left(E/w + (E/w)^{1-C_1} G(w) + w \log(E+2)\right)\Big).$$

With $\vartheta = 1/250$, its leading saving is

$$a_B = (1-\vartheta)C_1 + \vartheta a_L = \frac{859173879734039}{2500000000000000000}, \qquad \tau = 1 - a_B.$$

Exact rational comparison gives $a_B < \vartheta < 1 - a_B$, so the stopped term dominates the other root terms. The general reserve and finite profile are used by `prop:delayedbitcover`. The old `lem:threecoverstock` is the retained-stock template. Coefficient preparation is charged by `lem:rootdecoder`.

A.6 Diagonal invoices, the complement, and ordinary output

The four diagonal slots in a tensor round are product tables of already prepared local weights. If the prefix products have cardinalities $M_1, \dots, M_t$, their construction uses

$$\sum_{j=1}^{t} M_j < 2M$$

B-products: in reverse axis order these cardinalities form a geometric sum with ratio at most $1/2$. Applying the resulting table to all M sample positions is a separate M-product scan. Copies, inverses, erasure, and head returns are charged with those scans. An ordinary B-product costs $O_p(D \operatorname{polylog} L_0)$ by carry-free integer packing, classical integer multiplication, and reduction modulo the monic polynomial Φ_α [16, 17]. Coefficient slots have $O(\log D)$ bits; public integer-modulus reciprocals use $O(\log D)$ precision, while reversed monic reciprocals use their actual quotient precision. Thus building and applying one product-diagonal table costs

$$O_p(MD \operatorname{polylog} L_0),$$

not $O(Mt)$ ring products and not M fresh selector computations. The number of rounds is a fixed power of $\log R_0$, so all diagonal slots contribute $O_p(n(\log\log(n+4))^{K_2})$ for a fixed K_2.

It remains to account for the ψ-component. Write $d_\psi = \alpha - D$ and $v = d_\psi m = N\theta$. Classical Schönhage–Strassen arithmetic computes this component with inner stride $2d_\psi$ and long integer-modulus precision $O(L_0)$, so coefficient bands do not overlap and integer carries do not cross their slots. The full α-row monic division by ψ uses $D+1$ reciprocal coefficients; a short degree-d_ψ product reciprocal is not enough. Its total cost is

$$O_p(N\theta L_0^2 \operatorname{polylog} L_0) = O_p(N \operatorname{polylog} L_0)$$

by (1). All full coefficient-CRT row maps have the same $O_p(N \operatorname{polylog} L_0)$ bound. This complement is computed and recombined.

On the Φ_α component, embed the valid coefficient data into B, run two completed forward transforms, multiply pointwise in B, and run the inverse. The pointwise stage uses at most M ordinary B-products, hence costs

$$O_p(MD \operatorname{polylog} L_0) = O_p(n(\log\log(n+4))^{K_3})$$

for a fixed K_3, including the associated scans and head returns. Initially invalid dummy entries are zero and remain zero at every completed shear/route boundary; their point products are zero. The inverse uses the genuine m^{-1} on this reached valid subspace; no inverse assertion is made for arbitrary nonzero dummy fibres. Project away the restored helper coordinate, reconstruct both coefficient CRT components, invert the ordinary top map, crop only at its completed boundary, and output the $2n-1$ canonical residues. Since $N > 2n-1$, the zero-filled cyclic product has no aliasing among degrees 0 through $2n-2$. Input scans, output scans, and final cleanup cost $O_p(n)$.

Combining (9), the diagonal invoice, the complement, the coefficient maps, and the fixed polynomial setup proves the $e = 1$ case of the proposition. Each named supplier is a completed fixed-tape machine. Their finite bank sets are renamed and reused, and recursive setup stacks are tape data; joins, returns, and cleanup are included. No tape is added per axis, prime, depth, or gate.

Finally fix a basis $\{\beta_1, \dots, \beta_e\}$ of $\mathbb{F}_{p^e}$ over $\mathbb{F}_p$ and its structure constants $\beta_i\beta_j = \sum_k c_{ij,k}\beta_k$. A linear scan separates each input into its e prime-field component streams. Run the e^2 ordered component-pair products serially, reusing the same prime-field banks, then accumulate the fixed combinations $c_{ij,k}$ and merge the e output streams coefficient by coefficient. These

codecs and fixed linear combinations cost $O_{p,e}(n)$, and the e^2 calls are a constant factor. All $2n-1$ output coefficients, including zeros, are retained. This proves the fixed-extension case.

For the plain exponent, observe that

$$\frac{319}{320} - \gamma = \frac{805}{10^9} > 0.$$

For each fixed K, the factor $(\log\log(n+4))^K$ is absorbed by $(\log(n+2))^{805/10^9}$ for sufficiently large n; changing the constant covers the remaining finite range. This yields the stated 319/320 bound. □

Remark 1 (Source and verification boundary)**.** The native interfaces are `thm:mixedbitswap` and the phase-only `thm:mixedbitinteger` in the accompanying native supplement, with the constructive adapters and proof labels listed above. `ASSETS.json` binds the supplement and its finite premises; `SOURCE_SELECTION.json` records its derivation from the research source. Finite checks concern their stated finite identities and do not replace the all-size arguments. Complete Lean verification remains open.

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